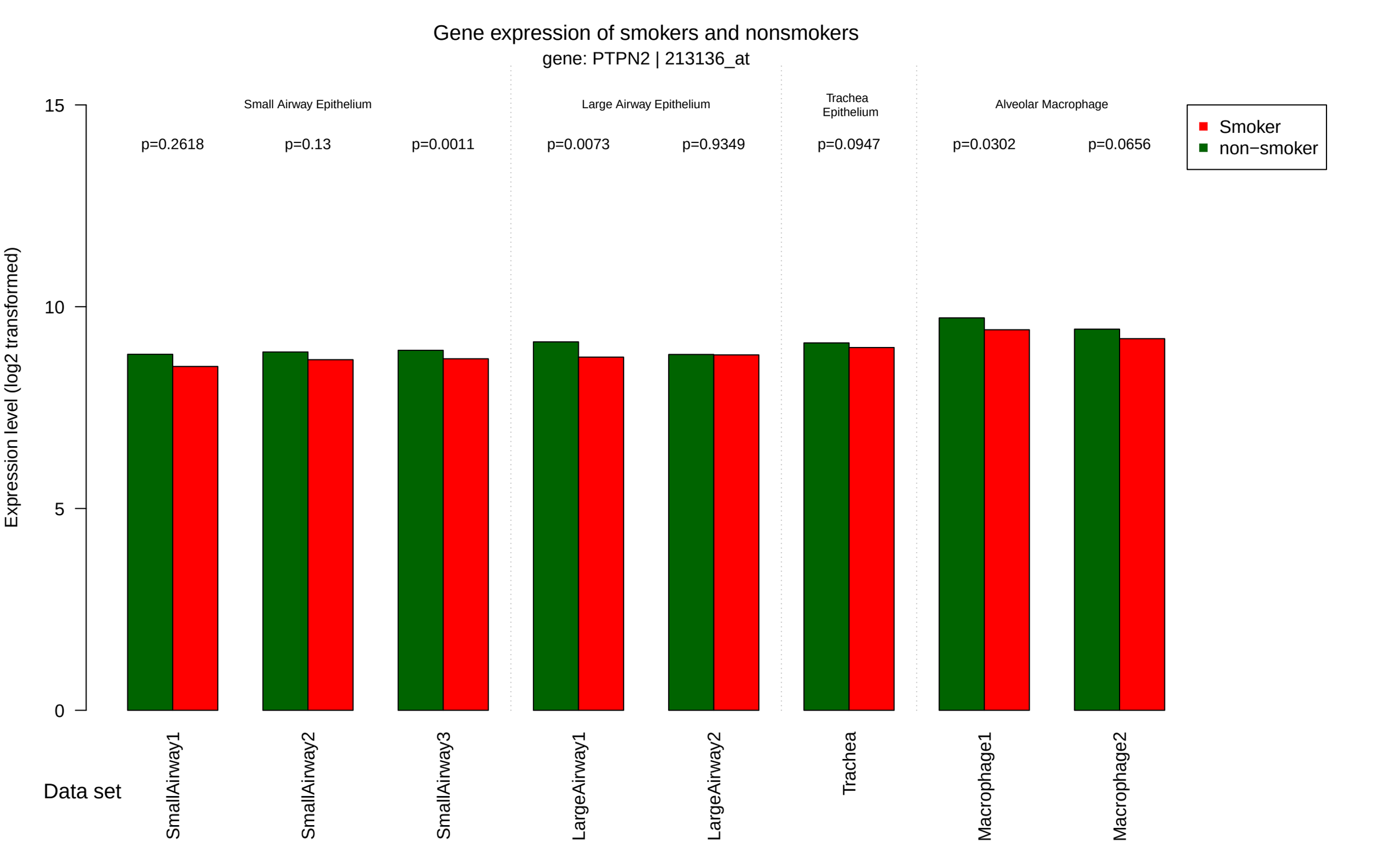
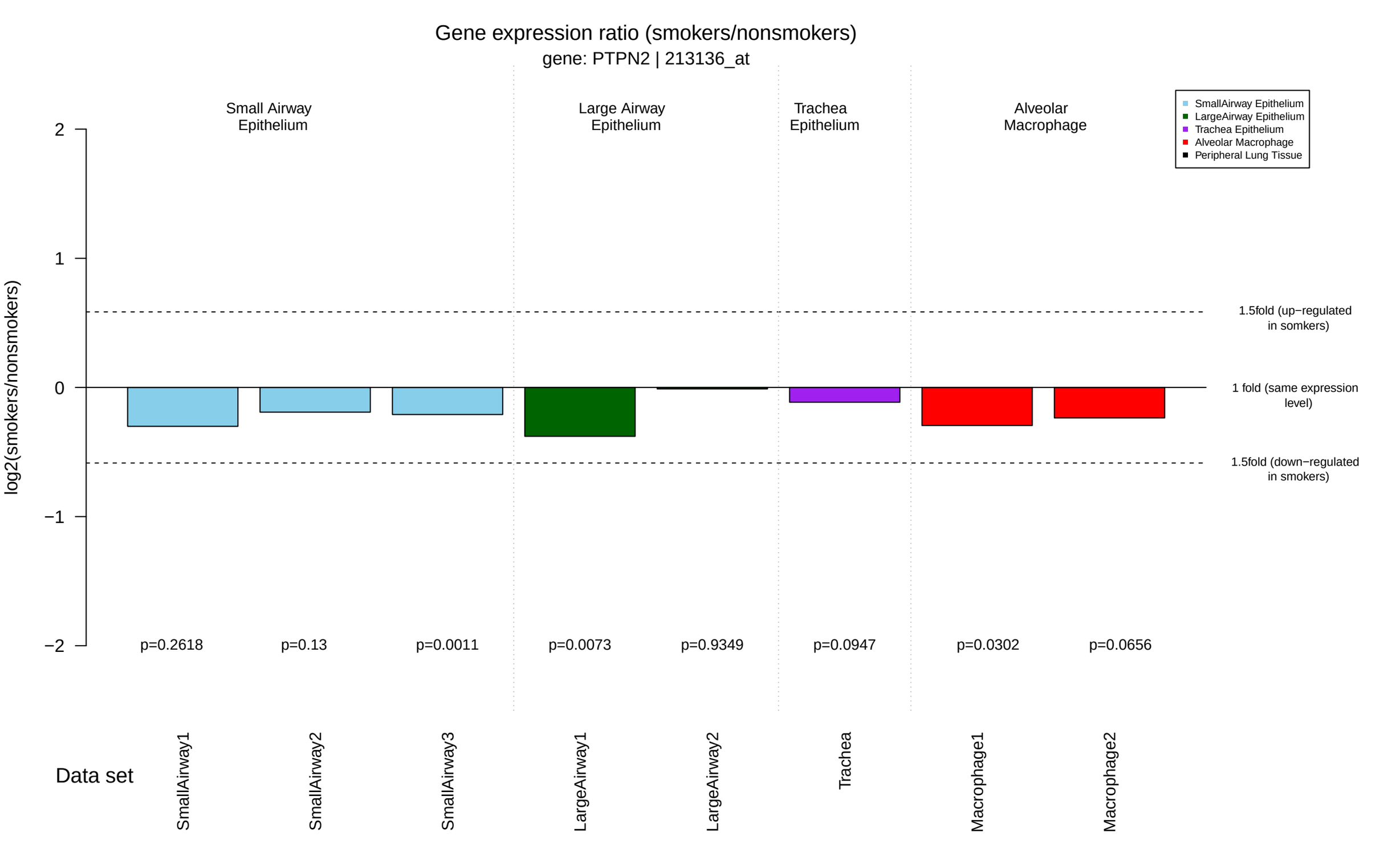
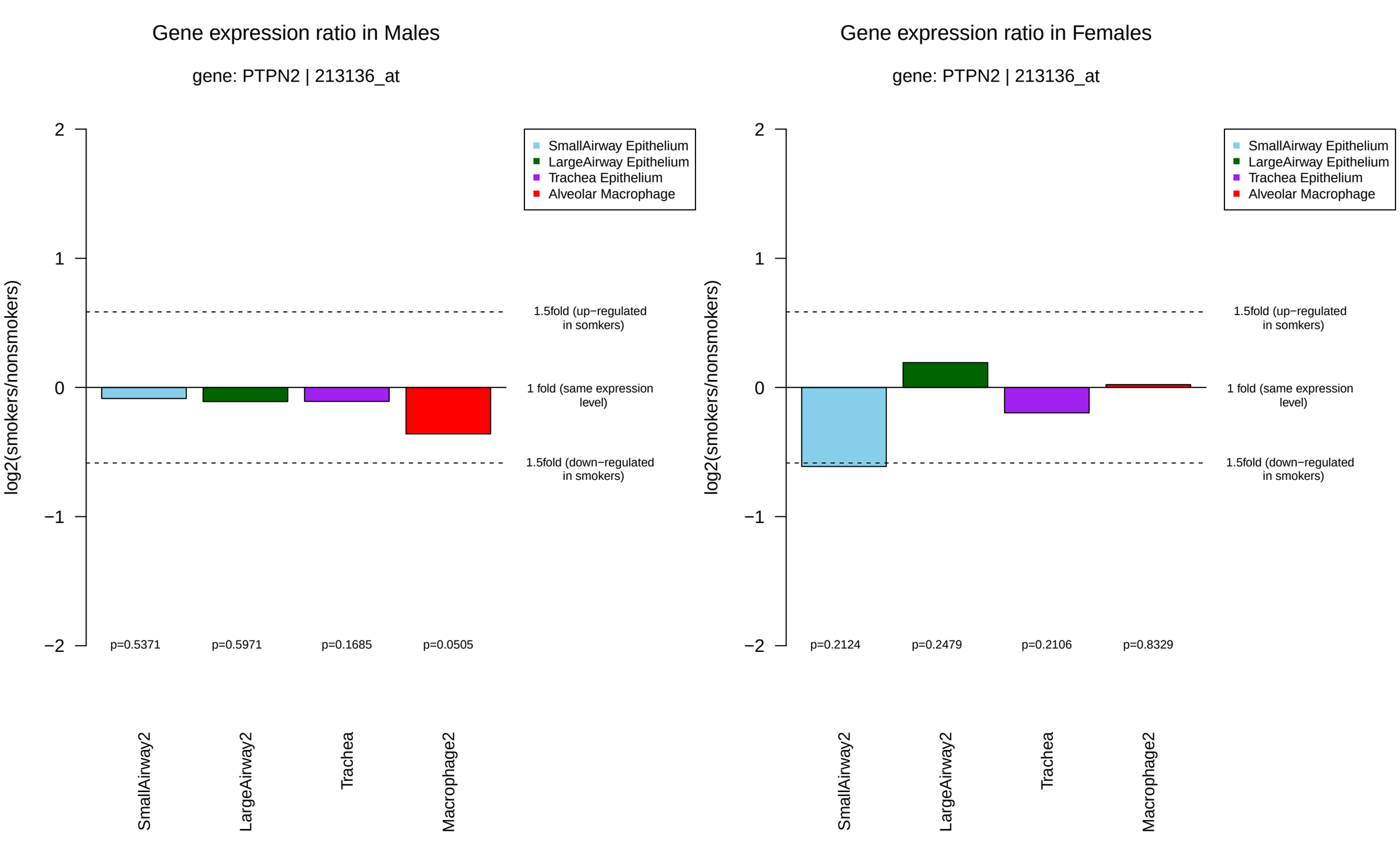
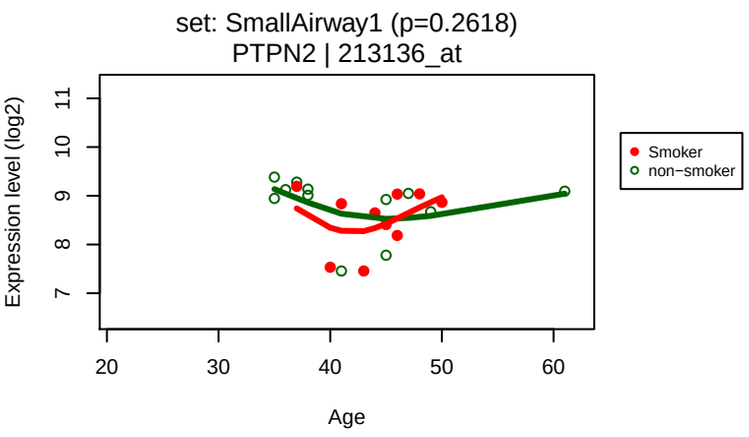
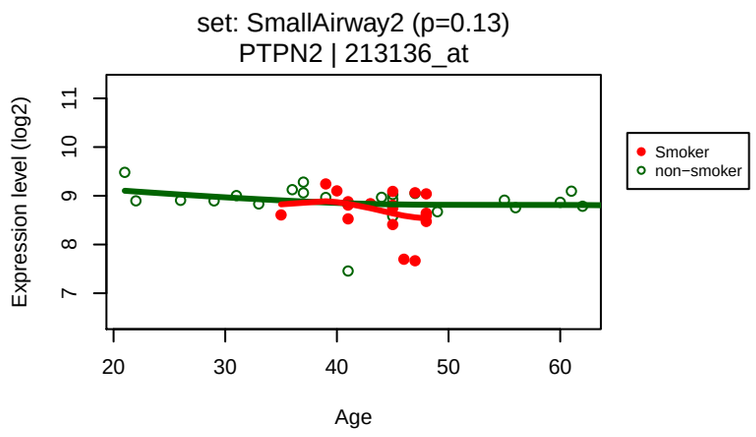
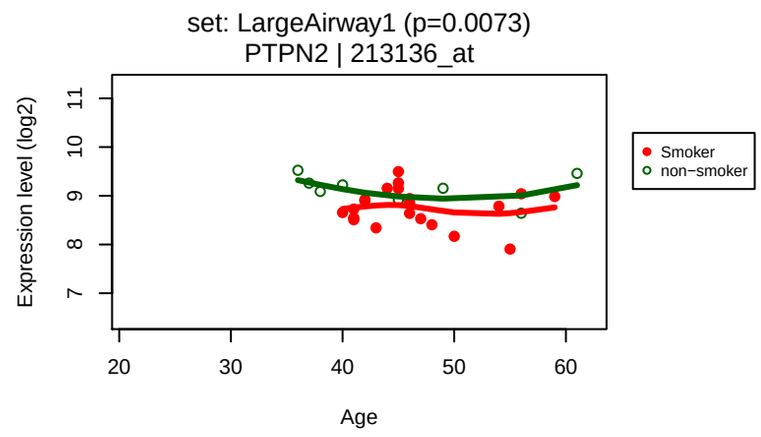

Gene expression of smokers and nonsmokers
gene: PTPN2 | 213136_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2618
p=0.13
p=0.0011
p=0.0073
p=0.9349
p=0.0947
p=0.0302
p=0.0656
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

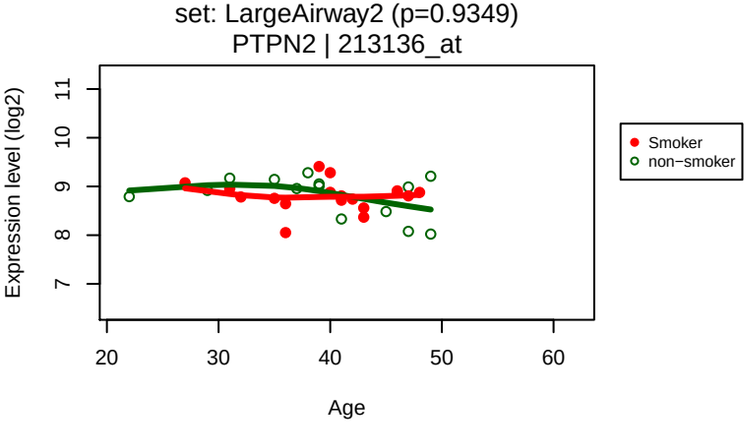

Gene expression ratio (smokers/nonsmokers)
gene: PTPN2 | 213136_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2618
p=0.13
p=0.0011
p=0.0073
p=0.9349
p=0.0947
p=0.0302
p=0.0656
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

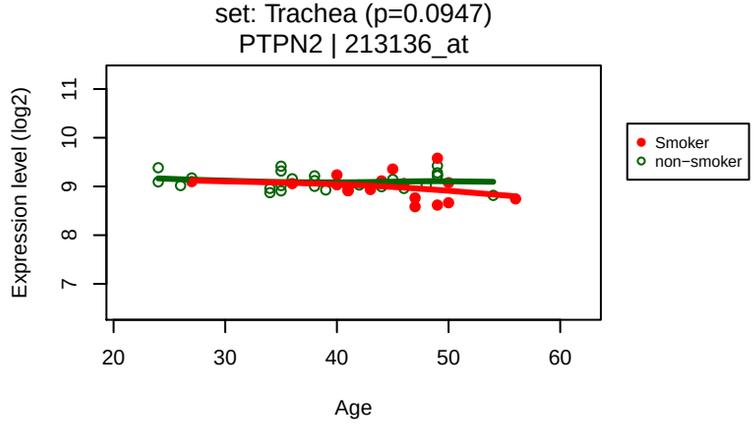

Gene expression ratio in Males
gene: PTPN2 | 213136_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5371
p=0.5971
p=0.1685
p=0.0505
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPN2 | 213136_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2124
p=0.2479
p=0.2106
p=0.8329
SmallAirway2
LargeAirway2
Trachea
Macrophage2

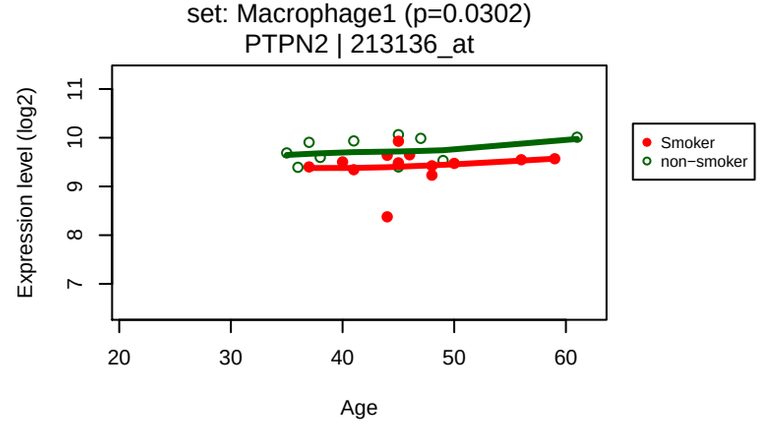

set: SmallAirway1 (p=0.2618)
PTPN2 | 213136_at
Expression level (log2)
Age
Smoker
non−smoker

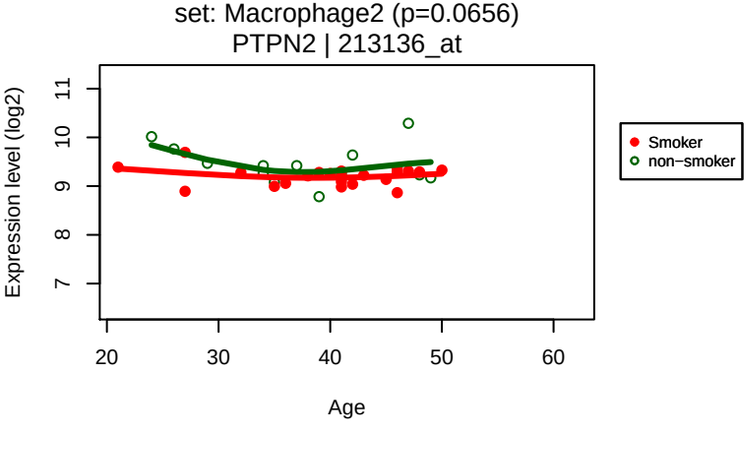

set: SmallAirway2 (p=0.13)
PTPN2 | 213136_at
Expression level (log2)
Age
Smoker
non−smoker

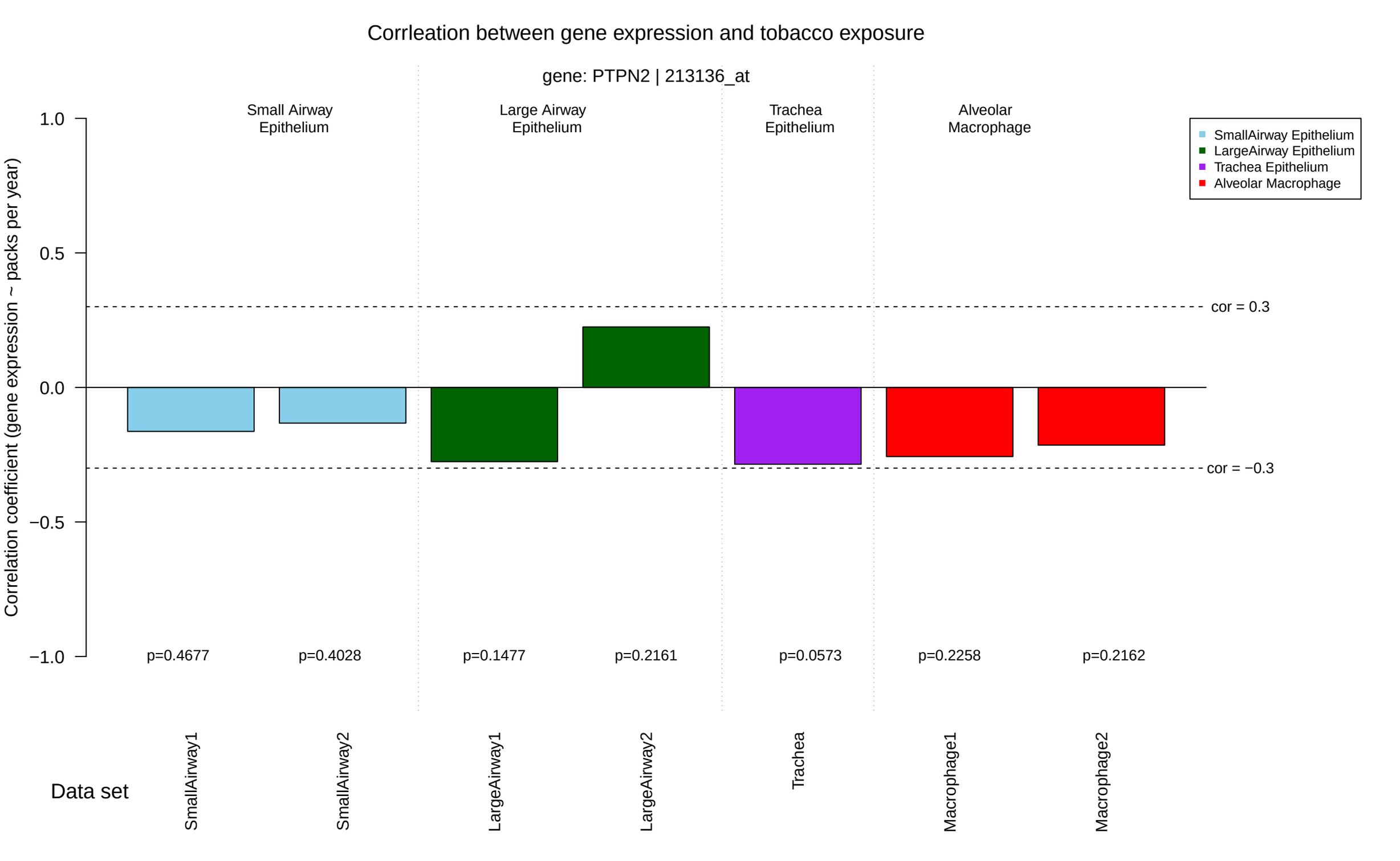

set: LargeAirway1 (p=0.0073)
PTPN2 | 213136_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9349)
PTPN2 | 213136_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0947)
PTPN2 | 213136_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0302)
PTPN2 | 213136_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0656)
PTPN2 | 213136_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPN2 | 213136_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4677
p=0.4028
p=0.1477
p=0.2161
p=0.0573
p=0.2258
p=0.2162
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2